Talkly - Technical Architecture Document

1. System Overview

Talkly is a comprehensive AI Voice Agent platform designed for managing outbound and inbound calls. It enables users to create campaigns, manage leads, configure AI agents, and process audio in real-time. The system comprises a modern web frontend and a robust backend processing engine.

2. Technology Stack

- Frontend: Next.js, React, Tailwind CSS
- Backend: FastAPI (Python), Uvicorn
- Database: MongoDB
- Audio Processing: FFmpeg, PyDub
- AI Integrations: OpenAI (GPT-4o), OpenRouter, Deepgram/ElevenLabs (for STT/TTS)
- Real-Time Communication: WebRTC / SIP integration

3. Architecture Breakdown - Backend (FastAPI)

- Auth Routes (/api/v1/auth): Handles user authentication, authorization, and multi-tenant access control.
- Campaign Routes (/api/v1): Management of calling campaigns, scheduling, and linking with agents/leads.
- Lead Routes (/api/v1): Ingestion, updating, and tracking of customer leads.
- Agent Routes (/api/v1): Configuration of AI agents, defining prompts, voice characteristics, and intelligence settings.
- Billing Routes (/api/v1/billing): Tracks usage and calculates costs based on the unit-wise breakdown (Rs 12/minute).
- Inbound Routes (/api/v1/inbound): Handling incoming calls and routing them to the appropriate AI agent context.
- Super Admin (/api/v1/admin): Platform-wide administration and analytics.

4. Services & Processing Engine

- Audio Processing (process_audio.py): Handles audio file uploads, chunking, and conversion. Integrates with Speech-to-Text APIs.
- Intelligence Service: Interfaces with LLMs to generate conversational responses.
- Language Service: Manages multi-lingual capabilities and translation configurations.
- Billing Service: Calculates deductions based on API usage (STT, LLM, TTS, Telecom).

5. Database (MongoDB)

MongoDB is used as the primary data store. The application uses document collections for:
- users, companies (Multi-tenant architecture)

- agents (Configurations for AI bots)
- campaigns & leads (Outbound marketing data)
- calls & transcripts (Call logs, analysis, and raw transcript storage)

6. Cost Structure & Unit Economics

The platform operates on a Rs 12 per minute unified charge which covers:
1. Telecom Connectivity: Rs 2.50
2. Real-Time STT: Rs 1.50
3. AI Processing (LLM): Rs 3.50
4. Text-to-Speech (TTS): Rs 2.50
5. Platform & Infrastructure: Rs 2.00

7. Security & Multi-Tenancy

- Tenant Isolation: All data operations are filtered by company_id to ensure strict tenant data isolation.
- Environment Management: API keys (OpenAI, OpenRouter) are securely managed via .env variables and are not exposed to the client.

8. Deployment & Infrastructure

- The backend is containerized using Dockerfile.
- FFmpeg is required on the host/container for media processing.
- The Next.js frontend is built statically or runs via Node.js server for SSR/ISR.